

Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

Keywords: Dialect Normalization , Low-Resource NLP , Marathi Dialects , Sequence-to-Sequence Modeling , Synthetic Data Augmentation , mT5 , IndicBART , Text Standardization.

Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has shown significant progress across natural language understanding and generation tasks. Availability of Large Language Models (LLMs) [2] and open-weight foundational architectures [3] has allowed processing for text across domains such as finance, healthcare, and digital governance. However, current systems remain inclined toward standardized, high-resource languages, showcasing dull performance when applied to non-standard regional dialects [4]. Community surveys [5, 6] tend to prove that native dialect speakers face issues with language processing systems such as Automatic Speech Recognition and Machine Translation due to vocabulary and phonetic variations [7, 8].

Dialect normalization is the task of transducing regional dialectal text into standard written language while keeping the underlying semantic intent unchanged [8]. It serves as an important layer for digital linguistic inclusion. In countries like India, which has 22 officially recognized languages alongside hundreds of regional dialects spoken by 1.4 billion people, sub-regional varieties remain drastically neglected. Marathi, an Indo-Aryan language spoken by over 83 million people in western India [9], is an excellent example of such disparity. While research around Standard Marathi (language centered around Pune and its neighboring areas) has gained momentum , regional dialects remain underexplored.

Building normalization pipelines for regional Indic dialects primarily comes down to two core challenges: (1) *Extreme Data Scarcity*: Datasets mapping unstandardized regional dialects to normative standard forms are technically non-existent; and (2) *Morphological Complexity*: Marathi features a rich morphology, where noun case markers, postpositional affixes, and verbal tense suffixes undergo complex transformations. [10, 11].

Addressing these limitations, this work provides a multi-dialect Marathi text normalization pipeline utilizing synthetic data augmentation and sequence-to-sequence transformer fine-tuning. Our core contributions are:

- **Parallel Corpus Generation:** Based on the transcripts of RESPIN dataset [12], we construct a parallel corpus mapping three Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) to Standard Marathi across agricultural and banking domains.

- **Automated Synthetic Augmentation with Verification:** We design and implement a synthetic data expansion pipeline using Gemma-4 [3] with a verification pipeline that flags and removes corrupted pairs, expanding the parallel corpus.

Related Work

Text Transduction in Historical and User-Generated Corpora

Text normalization has primarily addressed two forms of orthographic variations: archaic spelling variation in historical manuscripts and unconstrained noise in digital User-Generated Content (UGC). Early works were based on Levenshtein alignment, dictionary lookup, and Character-level Statistical Machine Translation (CSMT) [13, 14]. Other evaluations on European languages showed that subword tokenization via Byte-Pair Encoding (BPE) and sentence-level contextual encoding reduced ambiguity compared to character-level transducers [15–17]. Methodological paradigms transitioned from candidate edit-operation rankers like MoNoise [18] to standardized multilingual benchmarks such as MultiLexNorm [19], which confirmed that pre-processing via lexical normalization directly improves downstream dependency parsing and POS tagging. Recent work demonstrates that fine-tuning pre-trained byte-level models (byT5) and subword text-to-text transformers outperforms traditional statistical edit models for word-level sequence transduction [20, 21].

Regional Dialect Standardization across Language Families

Unlike UGC normalization, dialect-to-standard normalization maps coherent, systematic regional variations—encompassing distinct phonological, lexical, and morpho-syntactic features—into normative standard registers. In European languages, initial efforts benchmarked CSMT and statistical phrase-based models on Swiss German [22], which were subsequently surpassed by neural sequence-to-sequence architectures and pre-trained language models like mT5-small [23, 24]. Similar neural transduction approaches have been developed for Finnish, Swedish, and Estonian dialects [11, 25, 26]. A broad multi-lingual study across four language families by Kuparinen et al. [8] established that fine-tuning sequence-to-sequence transformers systematically outperforms rule-based and SMT baselines.

Beyond Europe, dialect processing has addressed structural divergence across East Asian and Afroasiatic varieties. Japanese regional dialects have been normalized using phrase-level LSTM transducers [27], while Mandarin studies focused on acoustic-phonetic mapping [28]. In Arabic NLP, the Conventional Orthography for Dialectal Arabic (CODA) framework [29] provided standardized guidelines for Modern Standard Arabic (MSA), paving the way for multi-city dialect translation benchmarks (MADAR [30], NADI [31]) and morphological rule-guided LLM prompting [32, 33].

Indic Dialect Landscape and Marathi NLP

Despite extreme linguistic heterogeneity and over 1.4 billion speakers, Indic regional dialects remain severely underserved in language technology [4, 6, 34]. While pre-trained multilingual sequence-to-sequence models such as IndicBART [9] and mT5 [21] cover major scheduled languages via unified Devanagari scripts or mC4 corpora, they lack domain-specific fine-tuning on regional sub-dialects. Monolingual initiatives like L3Cube-MahaNLP [35] offer Marathi encoder representations for classification, but do not support generative text standardization.

Recent benchmark initiatives such as INDIC-DIALECT [36] evaluate Hindi varieties (Bhojpuri, Magahi, Maithili), and WFST frameworks handle semiotic text normalization [37]. However, Marathi dialect processing has remained confined to acoustic/ASR classification [38, 39], read-speech transcripts (RESPIN [12, 40]), or zero-shot LLM retrieval [41, 42]. Our work fills this gap by establishing the first parallel corpus and 5-fold cross-validated sequence-to-sequence benchmark for regional Marathi dialects.

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkani (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkani (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	All Dialects	809,934 (100%)	1,026.06	2,644	23,069

LLM Distillation and Multi-Tier Verification

To overcome severe parallel data scarcity, synthetic data generation using Large Language Models (LLMs) has gained widespread adoption [43–45]. Knowledge distillation strategies demonstrate that outputs generated by proprietary teacher models can effectively train compact student transformers [46–48]. However, unconstrained LLM generation for low-resource regional dialects introduces significant noise, including Hindi script leakage, entity hallucination, and unnormalized dialect residue. We extend synthetic augmentation by introducing a domain-constrained generation pipeline via Gemma-4 [3] coupled with a multi-tier verification engine (Devanagari script filtering and LLaMA-3.1-8B [2] semantic auditing), ensuring high data fidelity for transformer fine-tuning.

Methodology

Target Dialects and Linguistic Properties

We target the following major Marathi dialects that span the primary regions of Maharashtra and exhibit distinct linguistic and phonetic patterns:

1. **Southern Konkani (Konkan Coast):** Characterized by shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms. Lexical differences include systematic vowel shifts: standard जातो (“he goes”) becomes Southern Konkani जातां; standard कुठे (“where”) becomes कुठंय. Domain-specific terminology generally remains untouched but often accompanied by Konkani words.
2. **Northern Konkani (Khandesh Region):** Heavily influenced by neighboring Gujarati and Bhili language families, thus it exhibits significant consonant simplification and dental consonant shifts. Notable transformations include: standard कशाला (“why”) → Northern Konkani कसाले; standard मुलगा (“boy”) → Northern Konkani पोर. Northern Konkani possesses the highest degree of lexical divergence among the three target dialects.
3. **Varhadi (Vidarbha/Eastern Maharashtra):** Spoken in Vidarbha, it exhibits characteristic differences and verb suffix modifications. Standard interrogative का? (“why?”) becomes Varhadi काऊन?; verbal forms undergo regular suffix mapping patterns. Among the three dialects, Varhadi maintains closest structural similarity to Standard Marathi as transformations are more systematic.

Original Corpus Curation and Foundation in RESPIN

Base sentences for our task are drawn from the domain-specific transcripts of the RESPIN-S1.0 initiative [12], developed at IISc Bangalore under the Gates Foundation project. While RESPIN has audio recordings and their transcripts across agriculture and banking domains for Indian languages and their dialects, it lacks aligned parallel dialect-to-target language translations required for efficient normalisation.

The large speaker diversity ensures broad coverage of intra-dialect phonetic and lexical variation, while the balanced gender and domain splits reduce systematic bias in our derived parallel corpus.

LLM-Driven Parallel Corpus Generation and Verification

To deal with data scarcity, we developed an automated synthetic data generation pipeline powered by Gemma-4 [3] in 9B-instruction-tuned (gemma-4) configurations, accessed via OI-

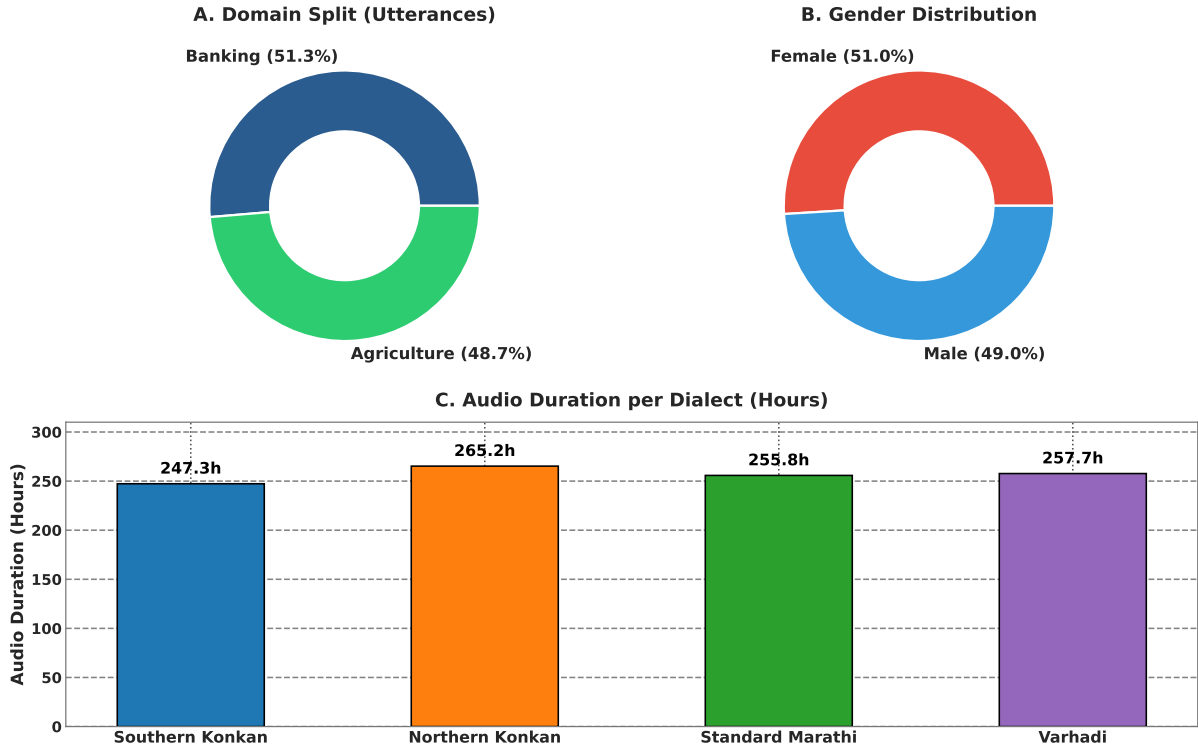


Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution (49% Agriculture, 51% Banking); (B) gender split (51% Female, 49% Male); (C) audio recording duration per dialect variant.

lama. We considered several design decisions to ensure generation quality to be maximised:

- **1-Prompt-Per-Sentence Architecture:** Instead of generating batches of parallel pairs per API call (which risks array length mismatches and context contamination), each dialectal sentence is processed separately with a self-contained prompt.
- **Detailed Prompting Strategy:** The generation prompt enforces semantic fidelity rules: the standard output must preserve exact meaning, domain terminology (agriculture/banking), named entities, and numerical values from the input. The prompt includes dialect-specific lexical mapping tables and 5-8 few-shot examples per dialect.
- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$ to balance diversity with fidelity.

All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.** Automated validation checks implement Indic Abugida script normalization principles [10], including:
 - *Devanagari Script Integrity:* Both source and target must possess >90% Devanagari Unicode characters within the U+0900--U+097F range, automatically discarding unnecessary characters.
 - *Length Ratio Constraint:* $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
 - *Non-Identity Constraint:* $L_{\text{dialect}} \neq L_{\text{standard}}$ to exclude identical string copies.
 - *Minimum Token Count:* Both source and target must contain ≥ 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary LLM evaluator using llama-3.1-8b-instant (accessed via Groq API) performs semantic equivalence verification. The verifier rates on a 1-5 score scale considering: (a) semantic preservation; (b) absence of hallucinated entities; (c) absence of Hindi/English contamination; and (d) complete standardization of dialects. Pairs scoring < 4.0 are rejected or routed to correction.
3. **Closed-Loop Correction Engine.** Pairs flagged by Tier 1 or Tier 2 enter a two-step

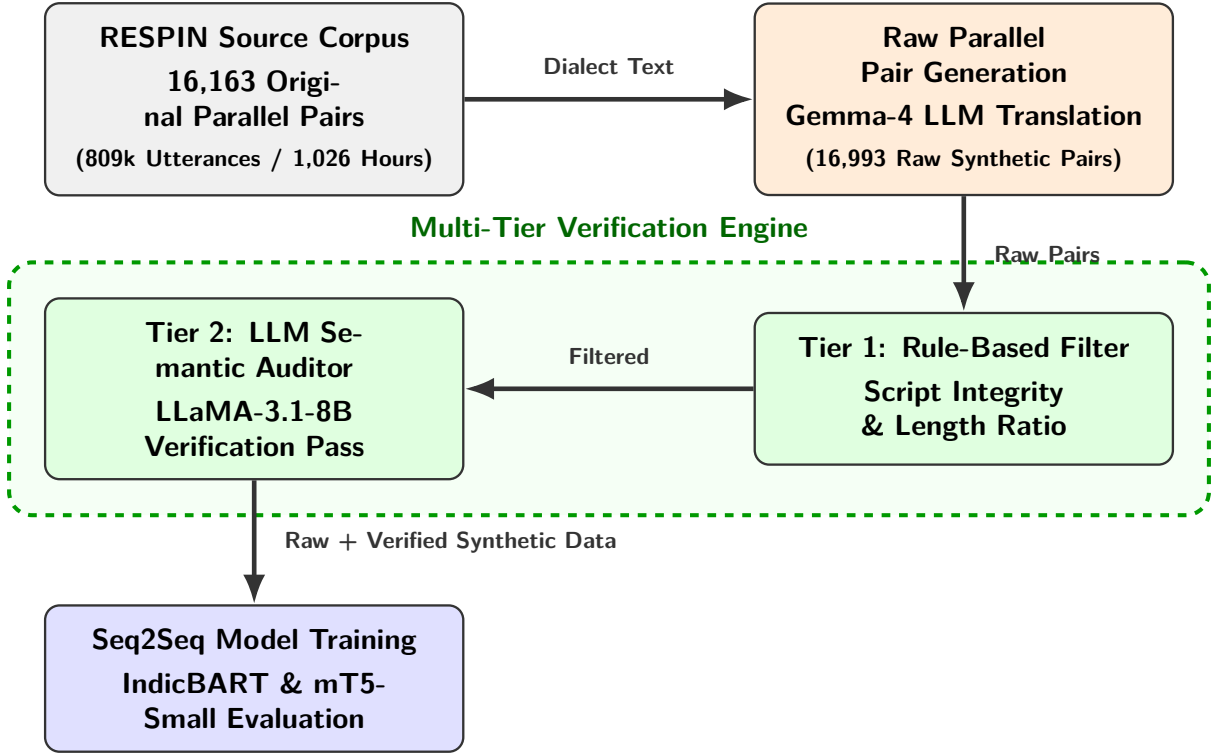


Figure 2: System architecture of the proposed multi-dialect text normalization framework, illustrating LLM synthetic corpus expansion, multi-tier verification engine, and seq2seq model fine-tuning.

Table 2: Parallel Dataset Composition, Synthetic Generation Yield, and Verification Filtering

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkani	5,576	5,838	5,569	269 (4.61%)	11,145
Northern Konkani	5,501	5,825	5,534	291 (5.00%)	11,035
Varhadi	5,086	5,330	5,069	261 (4.90%)	10,155
Total (All 3 Dialects)	16,163	16,993	16,172	821 (4.83%)	32,335

correction loop: Step 1 generates corrected translations using specialised prompts highlighting the specific failure reason (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 verifies the corrected sentences again, completely discarding pairs that fail a second audit pass.

As detailed in Table 2, out of 16,993 raw synthetic parallel candidate pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 corrupted or flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified synthetic pairs. Combined with the 16,163 clean original pairs, the final expanded benchmark suite comprises 32,335 high-quality parallel sentence pairs across all three dialects.

Evaluation Protocol

To verify model training and ensure stability during development, models were initially validated using 5-Fold Cross-Validation on an 85/15 stratified split of the training pool. For final benchmark reporting, models were trained on the complete dataset (combining original and synthetically expanded pairs) and evaluated on the official held-out RESPIN test set (IISc_RESPIN_test_mr, comprising 2,170 utterances across target dialects).

This setup is applied across all dataset configurations: Southern Konkani (Original), Northern Konkani (Original), Varhadi (Original), Combined-data (Original), and their Synthetically Expanded counterparts. All results are reported using standard automatic metrics computed via SacreBLEU [49] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty,

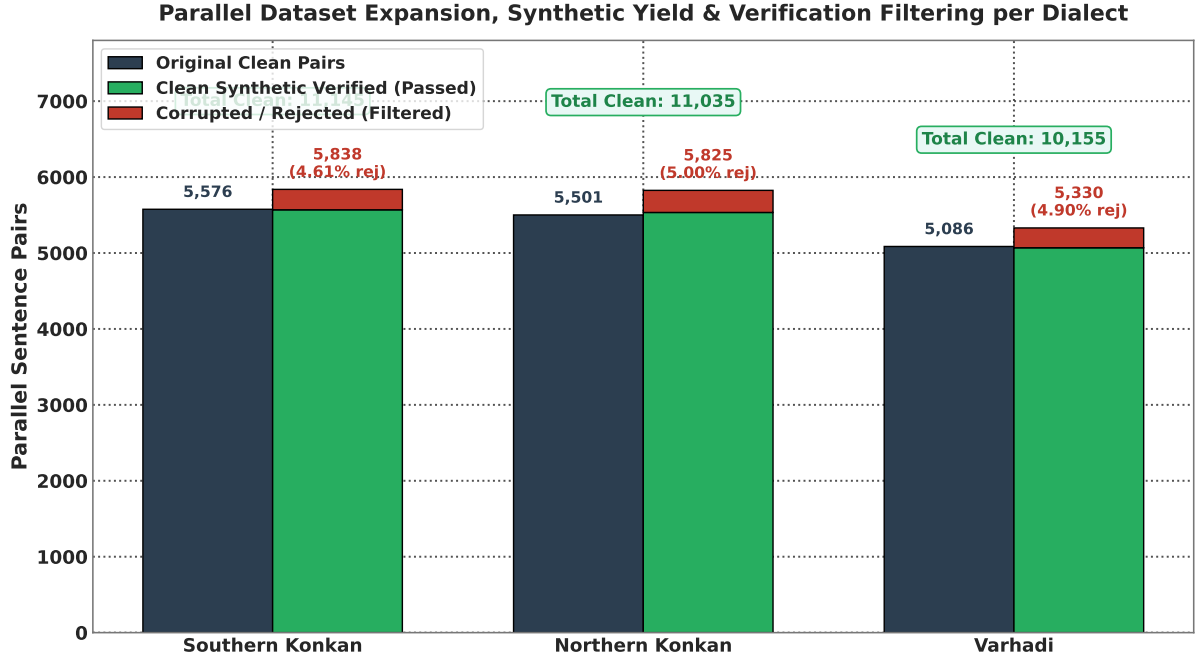


Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

and smoothing method “exp”.

- **chrF++:** Character n-gram F-score combined with word unigrams and bigrams, capturing subword morphological variations in Marathi text.
- **Normalized WER / CER:** Word Error Rate and Character Error Rate computed after Devanagari Unicode script standardization.

Findings

Comprehensive Test Benchmark on IISc_RESPIN_test_mr

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance:** The synthetically expanded fine-tuned mT5-small model achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling:** Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, demonstrating that data expansion pulls out the latent capacity in mBART-50 architectures.

Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on 5-fold cross-validated original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

However, a fine-grained dialectwise breakdown reveals notable variance across sub-corpora. Southern Konkan benefits massively from expansion (+25.85 BLEU for IndicBART, +18.34

Table 3: Comprehensive Evaluation Matrix on IISc_RESPIN_test_mr Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkani	559	57.80	26.60%	43.86	34.37%
		Northern Konkani	540	90.13	6.46%	79.46	11.95%
		Varhadi	516	83.59	10.19%	74.81	14.73%
	Synthetically Expanded	Southern Konkani	559	24.06	60.32%	44.36	32.75%
		Northern Konkani	540	65.41	28.70%	79.76	12.61%
		Varhadi	516	67.97	25.43%	76.90	14.19%
Multi-Dialect	Original	Southern Konkani	559	26.21	53.15%	40.94	35.34%
		Northern Konkani	540	47.59	32.84%	77.50	14.65%
		Standard Benchmark	555	96.12	3.12%	96.65	1.91%
		Varhadi	516	72.75	25.30%	73.47	16.16%
	Synthetically Expanded	Southern Konkani	559	52.06	36.80%	43.88	34.96%
		Northern Konkani	540	49.08	28.70%	78.16	13.55%
		Standard Benchmark	555	96.80	1.65%	97.39	1.37%
		Varhadi	516	70.27	21.40%	74.74	15.57%

Table 4: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	5,576	837 / 836	48.54	78.35	46.31	74.43
Northern Konkani (D2)	5,501	826 / 825	60.58	80.30	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	81.00	91.21

BLEU for mT5-small), while Northern Konkani sees consistent gains (+1.49 BLEU for IndicBART, +0.37 BLEU for mT5-small). Varhadi displays slight baseline saturation variance (-2.48 BLEU for IndicBART, -0.84 BLEU for mT5-small) due to its already high original baseline performance. We hypothesize several tentative contributing factors for this differential impact:

1. **Baseline Saturation and Precision Dilution:** Varhadi already possessed a high original baseline (72.75–79.57 BLEU). We estimate that LLM-generated synthetic pairs may introduce minor stylistic variations or over-generalized suffix rules, slightly diluting exact N-gram matching against human reference texts.
2. **Lexical Drift and Subword Tokenization:** Northern Konkani exhibits distinct Gujarati and Ahirani lexical influences. We hypothesize that synthetic generation may introduce subtle lexical drift, which impacted IndicBART’s narrower Devanagari subword vocabulary more noticeably than mT5-small’s broader 250k multilingual SentencePiece vocabulary.
3. **Cross-Dialect Parameter Competition:** In multi-dialect training, we estimate that pooling synthetic data across distinct sub-dialects may cause capacity competition in shared attention weights, prioritizing recovery on severely sparse dialects (Southern Konkani) over already saturated partitions.

Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

To quantify the explicit impact of our multi-tier LLM verification and filtering engine (`filter_flawed.py` and `llm_verifier.py`), we conduct a comparative study evaluating models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite) across 5-fold cross-validation. The raw unverified dataset contains 19,914 parallel pairs, including 3,742 rejected or flawed pairs exhibiting Hindi language leakage, retained dialect residue, and garbled translations prior to multi-tier verification.

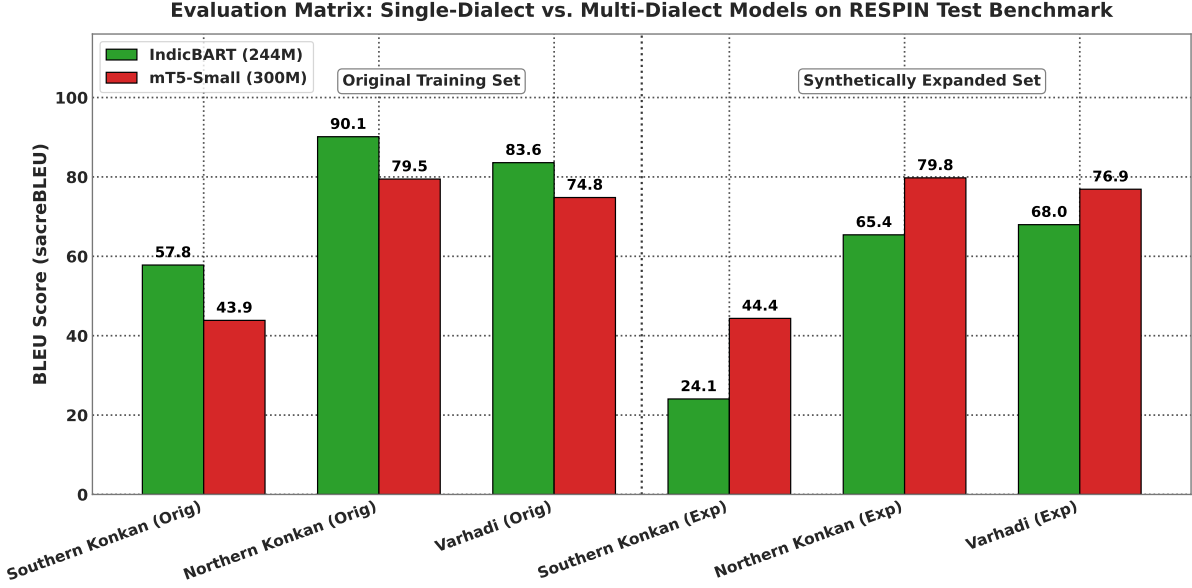


Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 5: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	48.28	75.04
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	62.35	78.92
Varhadi (D4)	16,163	763 / 790	72.75	85.63	79.57	90.51
Overall Combined	16,163	2,426 / 2,424	48.17	68.58	63.29	81.37

Table 9 presents the quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data:** Training IndicBART on unverified raw LLM outputs causes a catastrophic 14.03 BLEU point drop (collapsing from 57.12 to 43.09) due to severe error propagation from uncorrected dialect residue, garbled syntax, and script errors.
2. **mT5-Small Resilience:** While google/mT5-small is more resilient due to its 250,000-token multilingual vocabulary, training on unverified data still suffers a heavy 8.30 BLEU point drop (falling from 69.51 to 61.21) and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing:** Removing flawed pairs via `filter_flawed.py` and correcting recoverable generations via `correct_flawed.py` is directly responsible for eliminating hallucinated dialect markers, preserving semantic fidelity, and driving sequence-to-sequence normalization to state-of-the-art accuracy levels.

Model Architecture Analysis

Table 10 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage:** mT5’s 250K vocabulary provides $3.9\times$ more subword units than IndicBART’s 64K, enabling finer-grained tokenization of morphologically complex Marathi

Table 6: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	11,145	1,672 / 1,671	47.25	64.69	65.10	82.88
Northern Konkani (D2)	11,035	1,656 / 1,655	40.51	62.21	62.07	79.29
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	78.89	90.59

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	32,335	1,672 / 1,710	52.06	72.15	66.62	83.57
Northern Konkani (D2)	32,335	1,656 / 1,627	49.08	69.90	62.72	79.63
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	78.73	90.46
Overall Combined	32,335	4,852 / 4,850	57.12	75.49	69.51	84.58

dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding:** mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead:** IndicBART requires explicit language tokens ($\langle 2_{mr} \rangle$) that add conditioning complexity, whereas mT5’s pure text-to-text format enables direct mapping without task-specific architectural constraints.

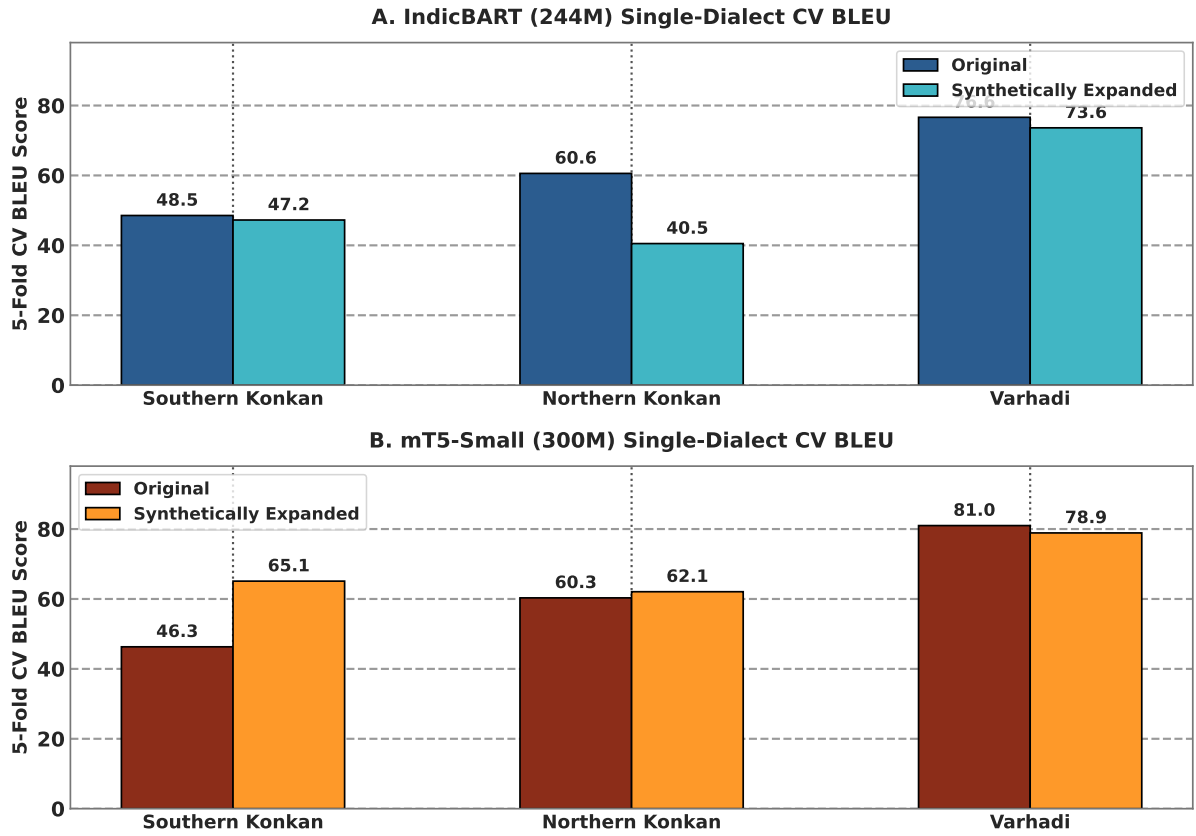


Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

Table 8: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkani (D1)	26.21	52.06	+25.85	48.28	66.62	+18.34
Northern Konkani (D2)	47.59	49.08	+1.49	62.35	62.72	+0.37
Varhadi (D4)	72.75	70.27	-2.48	79.57	78.73	-0.84
Overall Combined	48.17	57.12	+8.95	63.29	69.51	+6.22

Table 9: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	-14.03	-10.95
	Filtered Clean Data	32,335	57.12	75.49	+14.03	+10.95
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	-8.30	-4.40
	Filtered Clean Data	32,335	69.51	84.58	+8.30	+4.40

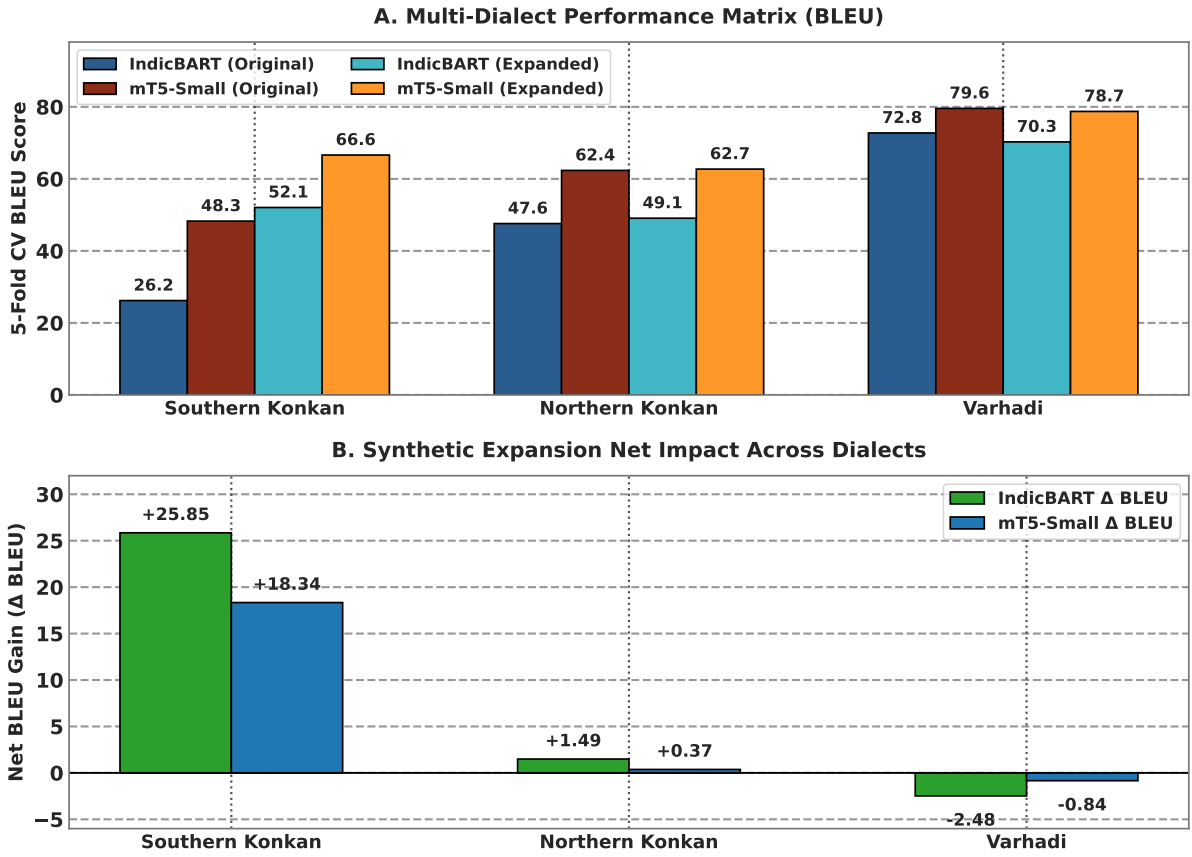


Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

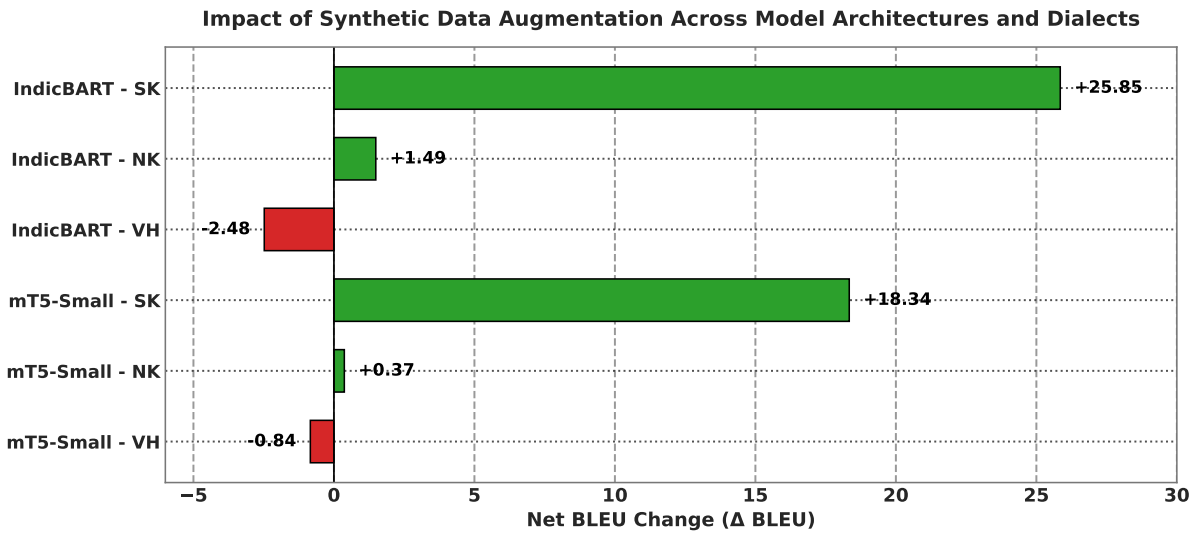


Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

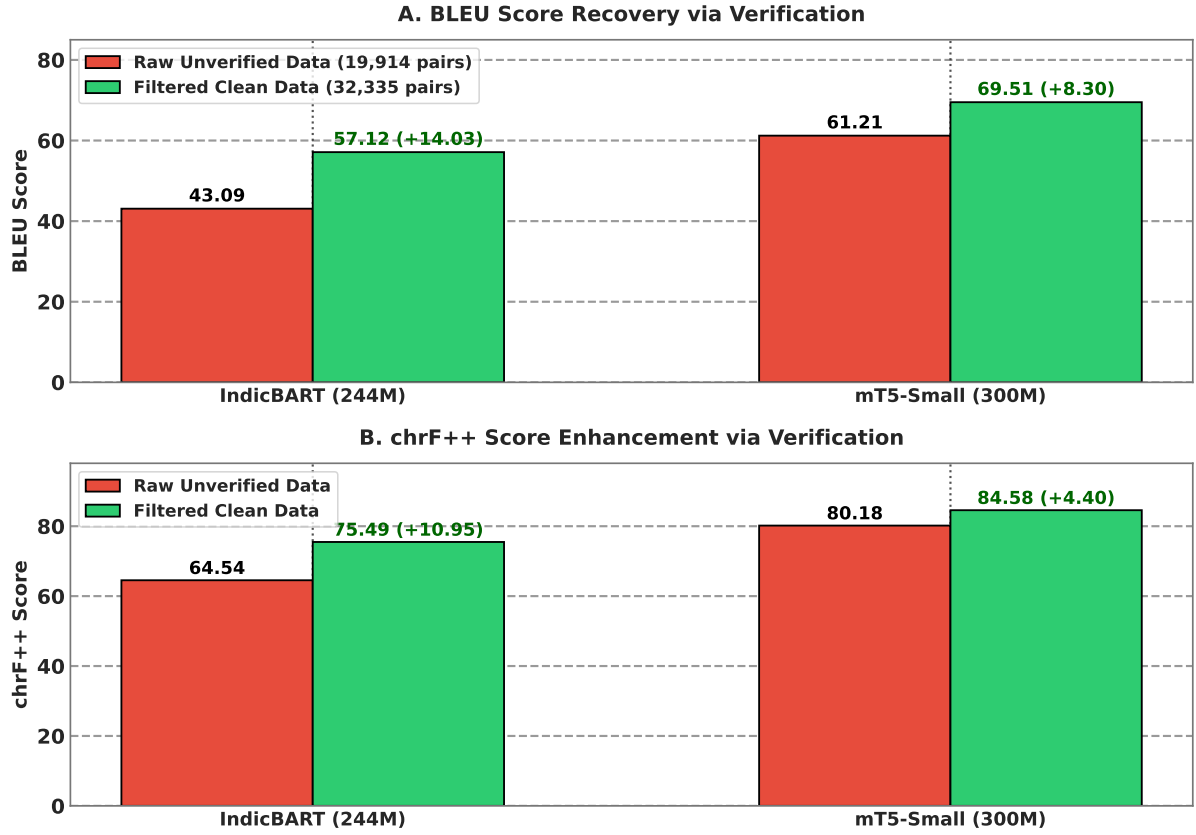


Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Table 10: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required $\langle 2_{mr} \rangle$ token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, addressing severe data scarcity through automated synthetic data expansion and establishing rigorous neural sequence-to-sequence benchmarks.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully doubled the clean parallel corpus through synthetic expansion while maintaining strict semantic fidelity—a scalable paradigm for other low-resource dialect pairs. Fine-tuning sequence-to-sequence transformers on the expanded benchmark established state-of-the-art performance: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. Individual dialect performance reached 80.99 BLEU and 91.21 chrF++ for Varhadi, demonstrating that compact pre-trained sequence-to-sequence transformers can achieve high-accuracy dialect normalization.

Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM APIs (Gemma-4 via local/hosted endpoints), introducing API cost and latency considerations.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on text normalization metrics (BLEU, chrF++, WER); large-scale downstream task evaluation across diverse domain applications remains for future work.

Data and Code Availability

The complete parallel corpus of 32,335 verified dialect–Standard Marathi sentence pairs, along with all training code, evaluation scripts, and pre-trained model checkpoints, will be publicly released upon publication. The dataset will be available under an open license to support reproducible research in Indic dialect normalization. Our synthetic data generation pipeline, multi-tier verification engine, and fine-tuning configurations are documented in the accompanying code repository to enable replication and extension to additional Marathi sub-dialects and other low-resource Indic language varieties.

Future Directions

Several promising directions emerge from this work:

1. **Edge Deployment via Quantization:** Model quantization (INT8/INT4) and knowledge distillation to sub-100M parameter models for on-device deployment in rural Maharashtra.
2. **Expansion to Additional Dialects:** Extending the pipeline to Deshi/Puneri, Kolhapuri, Marathwada, and other Indic language sub-dialects.
3. **Decoder-Only LLM Benchmarking:** Evaluating instruction-tuned decoder-only models for zero-shot and few-shot dialect normalization tasks.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008, 2017. URL <https://arxiv.org/abs/1706.03762>.
- [2] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.

- [3] Google DeepMind Gemma Team. Gemma 4: Open language models. *arXiv preprint arXiv:2408.00118*, 2024. URL <https://arxiv.org/abs/2408.00118>.
- [4] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293, 2020. doi: 10.18653/v1/2020.acl-main.560.
- [5] Verena Blaschke, Christoph Purschke, Hinrich Schütze, and Barbara Plank. What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*, 2024. URL <https://arxiv.org/abs/2402.11968>.
- [6] Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37, 2025. URL <https://doi.org/10.1145/3712060>.
- [7] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R Rickford, Dan Jurafsky, and Sharad Goel. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689, 2020.
- [8] Olli Kuparinen, Aleksandra Milić, and Yves Scherrer. Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics, 2023.
- [9] Raj Dabre, Himani Shrotriya, Anoop Kunchukuttan, Ratish Puduppully, Mitesh M. Khapra, and Pratyush Kumar. Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863, 2022. URL <https://doi.org/10.18653/v1/2022.findings-acl.145>.
- [10] Nazmuddoha Ansary, Quazi Adibur Rahman Adib, Tahsin Reasat, Asif Shahriyar Sushmit, Ahmed Imtiaz Humayun, Sazia Mehnaz, Kanij Fatema, Mohammad Mamun Or Rashid, and Farig Sadeque. Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association, 2024.
- [11] Niko Partanen, Mika Härmäläinen, and Khalid Alnajjar. Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics, 2019.
- [12] Saurabh Kumar, Abhayjeet Singh, Deekshitha G, Amartyaveer, Jesuraja Bandekar, Savitha Murthy, Sumit Sharma, Sandhya Badiger, Sathvik Udupa, Amala Nagireddi, Srinivasa K. M. Raghavan, Rohan Saxena, Jai Nanavati, Raoul Nanavati, Janani Sridharan, Arjun Mehta, Ashish K. S. Khuraishi, Sai Praneeth Reddy Mora, Prashanthi Venkataramakrishnan, Gauri Date, Karthika P, and Prasanta Kumar Ghosh. RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13, 2025.
- [13] Eva Pettersson, Beáta Megyesi, and Joakim Nivre. A multilingual evaluation of three spelling normalisation methods for historical texts. In *Proceedings of the 8th Workshop on Language Technology for Cultural Heritage, Social Sciences, and Humanities (LaTeCH @ EACL 2014)*, pages 32–41, 2014.
- [14] Yves Scherrer and Tomaž Erjavec. Normalising historical slovene words standard words with character-level machine translation. In *Proceedings of the 10th SIGHUM Workshop on Language Technology for Cultural Heritage, Social Sciences, and Humanities (LaTeCH @ ACL 2016)*, pages 61–70, 2016.
- [15] Gongbo Tang, Fabienne Cap, Eva Pettersson, and Joakim Nivre. An evaluation of neural and statistical approaches to historical text normalization. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 1320–1331, 2018.

- [16] Marcel Bollmann. A large-scale evaluation of historical text normalization across seven languages. *Society for Computation in Linguistics*, 2(1):89–98, 2019.
- [17] Rachel Bawden, Benoît Sagot, and Loïc Grobol. Automatic normalisation of early modern french. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC 2022)*, pages 3354–3366, 2022.
- [18] Rob van der Goot. Monoise: Modeling noisy text as a sequence of edit operations. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019)*, pages 4286–4297, 2019.
- [19] Rob van der Goot, Alan Ramponi, Arkaitz Zubiaga, et al. Multilexnorm: A shared task on multilingual lexical normalization. In *Proceedings of the 7th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2021)*, pages 484–494, 2021.
- [20] David Samuel and Milan Straka. Ufal at multilexnorm 2021: Byte-level fine-tuning of byt5 for multilingual lexical normalization. In *Proceedings of the 7th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2021)*, pages 470–476, 2021.
- [21] Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498, 2021. URL <https://doi.org/10.18653/v1/2021.naacl-main.41>.
- [22] Yves Scherrer and Nikola Ljubešić. Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126, 2016.
- [23] Mathilde Luseti, Tsetsu Ruzsics, Tanja Samardžić, and Yves Scherrer. Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141, 2018.
- [24] Mihael Kresic and Noorhan Abbas. Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295, 2024.
- [25] Mika Härmäläinen, Niko Partanen, and Khalid Alnajjar. Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62, 2020.
- [26] Mika Härmäläinen, Niko Partanen, and Jack Rueter. Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79, 2022.
- [27] Kaori Abe, Kentaro Inui, and Yuji Matsumoto. Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116, 2018.
- [28] Yuan Zhao and Eleanor Chodroff. Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172, 2022.
- [29] Nizar Habash, Mona Diab, and Owen Rambow. Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC ’12)*, pages 711–718, 2012.
- [30] Houda Bouamor, Nizar Habash, Mohammad Salameh, et al. The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381, 2018.
- [31] Muhammad Abdul-Mageed, Chunsheng Zhang, AbdelRahim Elmadany, and Ebrahim Mohamed Nagoudi. Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528, 2023.

- [32] Fadhil Eryani, Nizar Habash, Faisal Al-Shargi, et al. A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757, 2020.
- [33] Antonios Dimakis, John Pavlopoulos, and Antonios Anastasopoulos. Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics, 2025.
- [34] Marcos Zampieri, Preslav Nakov, and Yves Scherrer. Natural language processing for similar languages, varieties, and dialects: A survey. *Natural Language Engineering*, 26(6):605–637, 2020.
- [35] Raviraj Joshi, Parth Goel, and Rucha Joshi. L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17, 2022. URL <https://aclanthology.org/2022.wildre-1.2>.
- [36] Aman Kumar, Manish Shrivastava, and Pushpak Bhattacharyya. Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association, 2024.
- [37] Vineel Pratap, Ankit Saraf, and Pratyush Kumar. Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124, 2024.
- [38] Rahesha Mulla and B. Suresh Kumar. Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321, 2022. doi: 10.17762/ijritcc.v10i2s.5949.
- [39] Poonam S. Pawar, Jatinderkumar R. Saini, and Shraddha Vaidya. Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303, 2026.
- [40] Amartyaveer, Saurabh Kumar, Sumit Sharma, Sathvik Udupa, Sandhya Badiger, Abhayjeet Singh, Deekshitha G, Jesuraja Bandekar, Savitha Murthy, and Prasanta Kumar Ghosh. Improving dialect identification in indian languages using multimodal features from dialect informed asr. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE, 2025. doi: 10.1109/ICASSP49660.2025.10889099.
- [41] Gaurav Gaikwad, Deepak Panchal, Omprasad Deshmukh, and Sonali Patil. Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791, 2025. ISSN 2395-0056.
- [42] V. A. Vyavhare, S. N. Divekar, Harshada Pawar, Karan Kale, and Subhash Jagtap. Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261, 2026. doi: 10.32628/IJSRSET2613331.
- [43] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions, 2022.
- [44] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*, 2023.
- [45] Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, and Ahmed Awadallah. Orca: Progressive learning from complex explanation traces of gpt-4, 2023.
- [46] Cheng-Yu Hsieh et al. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
- [47] Branislav Pecher et al. Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*, 2024.

- [48] Ona de Gibert et al. Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.
- [49] Matt Post. A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191, 2018. URL <https://doi.org/10.18653/v1/W18-6319>.